

RAPHAEL HUANG – CURRICULUM VITAE

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RESEARCH INTERESTS

FPGA design, FPGA security; Heterogenous computing for ML; VLSI, place and route algorithms; Formal verification of hardware systems; Connected and automated vehicles, modular robotics.

EDUCATION

University of California, Santa Cruz September 2025 - Present

Master of Science in Computer Science and Engineering

Expected Graduation: December 2026

University of California, Santa Cruz September 2021 - June 2025

Bachelor of Science, *magna cum laude* in Computer Engineering

GPA: 3.93

PROFESSIONAL EXPERIENCE

Graduate Teaching Assistant, UC Santa Cruz, Baskin Engineering September 2025 - Present

- Coordinated logistical and technical aspects of a ~150-student Introduction to Logic Design course.
- Piloted the course's transition to a new open-source hardware toolchain, improving equitability.
- Oversaw the creation of homework and exams, ensuring consistent evaluation of student learning.
- Delivered occasional lectures and provided one-on-one assistance to students in lab sections, helping to debug large structural Verilog programs in Vivado, and clarify complex concepts.

Connected and Automated Vehicle R&D Intern, Argonne National Laboratory June 2025 - August 2025

- Transitioned LARS (Longitudinal Automated Replay System), a real-time vehicle energy analytics tool, from Simulink to a modular ROS 2 stack, enabling rapid cross-vehicle deployment.
- Validated the system through simulation and on-road testing on a Cadillac LYRIQ, implementing an Iterative Learning Control algorithm that reduced root-mean-square tracking error by ~65%.
- Delivered an accessible platform to support research and workforce development for the collegiate EcoCAR EV Challenge and the broader CAV community [R2].

Instructional Design Research Assistant, UC Santa Cruz, Hardware Systems Collective Jan - July 2025

- Redesigned the curriculum for CSE 100, Introduction to Logic Design, focusing on accessible, equitable, and pedagogically-sound coursework.
- Created homeworks and labs that enabled remote completion of coursework that enabled course enrollment to scale to over 140 students per quarter (previously ~80).

Lead Instructor, iDTech Camps, Stanford University July - August 2024

- Led 11 instructors teaching AI/ML in Python and VEX Robotics to 100+ students while adapting curriculum for a pilot program for international learners from China and Latin America.

RESEARCH AND PROJECTS

Runtime Reconfigurable LUT-based Logical Neural Networks on Zynq January 2026 - Present

- Developed a nanosecond scale inference architecture on Xilinx Zynq SoCs by mapping differentiable lookup table neural networks to reconfigurable FPGA overlays, eliminating traditional synthesis and bitstream loading bottlenecks.

Characterizing Sensor Accuracy of FPGA Side-Channel Attacks January 2026 - Present

- Constructing and evaluating the granularity of several voltage fluctuation sensor implementations and how they influence the statistical accuracy of FPGA power side-channel attacks.

Formal Methods for VLSI Timing Optimization September 2025 - January 2026

- Developing a formal methods approach to correct Static Timing Analysis (STA) violations by integrating a Z3 OMT solver with the OpenROAD open-source ASIC design flow.
- Created Python and Tcl scripts to parse timing reports, generate logical constraints for optimal cell resizing, and programmatically modify netlists [R1].

Vivaldi: A SystemVerilog FPGA Audio Synthesizer January - March 2025

- Wrote large-scale RTL in SystemVerilog for a Artix-7 FPGA to implement a multi-oscillator audio synthesizer; automated the verification and synthesis flow and validated the design with a comprehensive DPI-C testbench.

RISC-V on PYNQ September 2024 - March 2025

- Updated a tutorial that uses Python Jupyter notebooks to teach the user how to create and package a RISC-V processor overlay for the Xilinx PYNQ-Z2 board.

FPGA UART ALU January 2025

- Developed a custom variable-length packet parsing state machine to drive a 32-bit ALU via UART, enabling real-time verification with Python.

RELEVANT COURSEWORK

Graduate (M.S. in CSE)

- Formal Methods (CSE 216)
- Advanced VLSI Design (CSE 222A)
- Computer Architecture (CSE 220)
- Brain-Inspired ML (CSE 210)

Undergraduate (B.S. in CE)

- Logic Design with Verilog (CSE 125)
- Embedded System Design (CSE 121)
- Probability & Statistics (CSE 107)
- Signals & Systems (ECE 103)

HONORS, AWARDS, AND FELLOWSHIPS

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| [A4] | 2026-2027 | Graduate Pedagogy Fellowship, Teaching and Learning Center, UC Santa Cruz |
| [A3] | 2025 | Impact Argonne Award: Energy Secretary Visit, Argonne National Lab |
| [A1] | 2021 | UC Regents Scholar, Regents of the University of California |

TECHNICAL REPORTS

- [R1] C. Baker, R. Huang, W. Ali, A. Muddamsetti, *Optimal Cell Resizing in OpenROAD-flow-scripts Using Optimization Modulo Theories*, Baskin School of Engineering, University of California, Santa Cruz, Santa Cruz, California, 2025.
- [R2] Goberville, Nick and Huang, Raphael, *Modernizing LARS: Open-source, Modular, and Programmatically Flexible Scenario Replay*, Argonne National Lab, Science Undergraduate Laboratory Internships (SULI), Lemont, Illinois, 2025.
- [R3] R. Huang, D. Lee, C. Baker, V. Harkins, and G. Austin, *PillPal: A Smart Medication Dispensing System*, Baskin School of Engineering, University of California, Santa Cruz, Santa Cruz, CA, 2025.